

Adsorption Kinetics

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Abstract

Adsorption is a general phenomenon that allows the saturation of the force field generated by partially unsaturated surface atoms. The nature of the forces involved are physisorption (for van der Waals forces) or chemisorption (for chemical forces). Physisorption phenomena involve lower energies than chemisorption phenomena. The major difference is related to the time required to reach equilibrium.

INTRODUCTION

Adsorption as a Way to Restore Equilibrium

Any finite portion of matter is limited by a surface. Under equilibrium condition any body has a characteristic regular shape, like the sphere for isotropic fluids in the absence of gravity or a Wulff polyhedron for unisotropic solids. Because bulk atoms or molecules are arranged in the most convenient way to attain a configuration of minimum free energy, surface atoms are therefore subjected to unsaturated chemical or van der Waals forces, the degree of insaturation increasing with the dispersion of the solid. The surface free-energy excess of the system may be partially relieved via a number of phenomena like relaxation, reconstruction, or (in the presence of an atmosphere) *adsorption*. Adsorption is therefore a general phenomenon that allows the saturation of the force field generated by partially unsaturated surface atoms. According to the nature of the forces involved in adsorption, one speaks of *physisorption* (for van der Waals forces) or *chemisorption* (for chemical forces). These phenomena usually involve different energies and are manifested in different temperature ranges. Irrespective of the nature of the considered atoms, physisorption phenomena involve lower energies than chemisorption phenomena; the border between these domains is around 0.3 eV. Though both chemical and van der Waals forces are short range in character, they however differ because the former display saturation while the latter are, to a first order, pairwise additive. In spite of this difference, adsorption equilibrium and kinetics can to a large extent be described without explicitly considering the nature of the forces involved. The major difference is related to the time required to reach equilibrium.

In most cases, physisorption is not a thermally activated process, so that it occurs even at very low

temperature. In contrast, chemisorption is often a thermally activated phenomenon, so it may be seen only at elevated temperature. The absence of any activation energy for physisorption allows the fast attainment of equilibrium conditions, so the equilibrium isotherm is usually easily accessible to experiment. Of chemisorption one usually discerns kinetics only.

Adsorption–desorption phenomena occur in the following cases: gas at a solid surface, gas at a liquid surface, liquid at the liquid–solid interface, and liquid at the (immiscible) liquid–liquid interface. The most simple situation to rationalize is by far the first one, though much of what holds for it can be extended to the other situations.

EXPERIMENTAL DETERMINATION OF KINETICS OF ADSORPTION AND DESORPTION

The experimental determination of the adsorption or desorption rate, $J^{\pm}$ (J^{+} for adsorption and J^{-} for desorption), as a function of time, t , may be carried out with a lot of different experimental techniques. For the sake of concreteness, a particular technique (the gas-flow technique) is described here.

The Gas-Flow Technique

The gas-flow technique requires an experimental arrangement like that sketched in Fig. 1. The adsorbent, A, is contained in a vessel, V, of volume V connected to the external world through an orifice, O, of known geometry. The pressure of the adsorbate gas inside the vessel, P , and outside the vessel in the proximity of the orifice, P_m , are measured through suitable (e.g., Pirani or Bayard-Alpert) pressure gauges, G and G_m , respectively. The pressure P_m outside the system can be

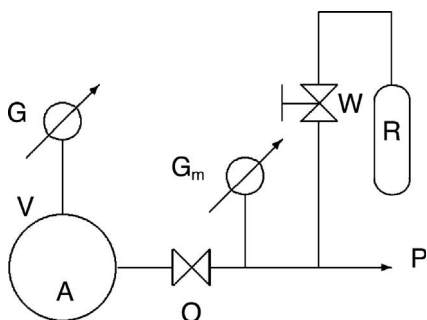


Fig. 1 Sketch of the apparatus for the gas-flow determination of adsorption-desorption kinetics.

controlled by opening a valve, W, separating the adsorbate reservoir, R, from the external side of O. The system is evacuated by a pump, P. The nature of the pumping system, constitutive materials, gaskets, etc., combined with the experimental protocol (maximum baking temperature before adsorption, etc.), determine the ultimate pressure inside the system. Pressures as low as 10^{-9} Pa are now achievable with standard ultra-high-vacuum techniques. This arrangement can be used for the determination of adsorption and desorption kinetics, using different protocols.

A protocol for adsorption kinetics

The protocol for the determination of the adsorption kinetics, based on keeping the adsorbate at a constant pressure p , requires that the external pressure p_m is progressively increased as adsorption proceeds. This is usually done by feeding back the difference between the actual pressure p measured by G and the goal pressure, p_g , to the valve W, which is progressively opened so as to keep the difference $|p - p_g|$ as close as possible to 0. The adsorbate temperature, T , is usually kept at a constant value, which in many cases may be assumed to be the same as that of the gas in the vessel V.

At any given time t the adsorption rate J^+ is determined by the pressures p_m and p ; in particular, at sufficiently low pressure the flux is proportional to the pressure difference:

$$J^+(t) = F[p_m(t) - p], \quad (1)$$

where F , the conductance of the orifice, depends on its geometry and the molecular mass of the gas. Eq. 1 is usually referred to as the Knudsen law; for its derivation, see Clausing.^[1] The total adsorbed amount, Q^+ , is thus given by

$$Q^+(t) = \int_0^t J^+(t) dt. \quad (2)$$

Eliminating t between Eqs. 1 and 2, we get how J^+ depends on the adsorbed amount Q^+ , $\tilde{J}^+(Q^+)$. The

functions $J^+(t)$, $Q^+(t)$, and $\tilde{J}^+(Q^+)$ are all representative of the adsorption kinetics. Because they are obtained under isobaric and isothermal conditions, they should therefore be denoted with the indices T and p . Varying either T or p , one can obtain the family $\{J_{T,p}^+(t)\}$ and the related ones, $\{Q_{T,p}^+(t)\}$ and $\{\tilde{J}_{T,p}^+(Q^+)\}$.

A protocol for desorption kinetics

The protocol for the determination of the desorption kinetics is based on keeping the external pressure p_m to values as low as possible, and in controlling the adsorbate temperature in such a way that the internal pressure p remains so low as to give a negligible re-adsorption. Combining conservation of matter with the gas law, one has

$$J^- = \frac{V}{k_B T} \frac{dp(t)}{dt} + F[p(t) - p_m], \quad (3)$$

where k_B is the Boltzmann constant and the gas has been supposed so rarified as to allow its flux through O to be described by the Knudsen law. In general, $p_m \ll p(t)$, so that $J^-(t)$ may be determined by recording $p(t)$. Once $J^-(t)$ is known, $Q^-(t)$ is immediately calculated by an integration, and $\tilde{J}^-(Q^-)$ is obtained eliminating t from $J^-(t)$ and $Q^-(t)$.

Combining the Gas-Flow Technique with Other Techniques

The apparatus sketched in Fig. 1 is a kind of absolute minimum for the determination of the adsorption-desorption kinetics. In fact, it determines both the rate and amount by measuring pressures only, keeping the system at constant temperature. The quality and reliability of the datum can, however, be improved using the apparatus in combination with others; for instance, the adsorbed amount can be measured independently by means of a microbalance, the gas composition can be monitored with a mass spectrometer, the gas uniformity inside the vessel can be checked by using a few pressure gauges, the heat of adsorption may be determined via microcalorimetry, the adsorbent temperature can be varied with time in a programmed way, etc.

NONIDEALITIES OF REAL ADSORBENTS

The adsorbents of major practical interest are highly dispersed. They may be solid (powders) or liquid (foams), amorphous (carbon, silica, etc.) or crystalline (zeolites, clays, etc.), conductive (metals) or insulating (oxides, sulfides, etc.), organic or inorganic; *their unique common feature seems to be their exceedingly high surface area*, as measured by standard techniques like the

ones based on the analysis of N₂ adsorption isotherms (see, for instance, Young and Crowell^[2]). Silica materials with surface area greater than 10³ m²g⁻¹ (in which each Si(O-)₄ tetrahedron constituting the material is exposed to the atmosphere) are easily prepared, for instance, via the sol-gel route.

Energy Heterogeneity

In view of their high degree of dispersion, the surfaces of real adsorbents are characterized by a large number of faces, edges, and vertices. This situation implies the existence of a spectrum of adsorption energies. The adsorption energy q here considered is the one determined in the limit of vanishing coverage. The larger the dispersion degree of the adsorbent, the wider and denser is the spectrum D_q of the adsorption energy q . For sufficiently dispersed adsorbents, the spectrum may be described by a continuous function, $\phi(q)$, where $\phi(q)dq$ represents the fraction of adsorbent with adsorption energy between q and $q + dq$. By definition, $\phi(q)$ is nonnegative,

$$\forall q(\phi(q) \geq 0), \quad (4)$$

and normalized,

$$\int_{D_q} \phi(q)dq = 1. \quad (5)$$

In many cases, the integration limits in Eq. 5 are taken to be 0 and $+\infty$, remembering, however, that the support of $\phi(q)$ is contained in the interval (q_m, q_M) , where q_m and q_M denote the infimum and supremum, respectively, of the adsorption-energy spectrum.

While in the zero-coverage limit the topographic structure of the surface does not affect the adsorption kinetics, it becomes important at finite coverage. In particular, the following limiting cases seem to cover most of the situations that are observed in practice: quasi-homogeneity, patchwise heterogeneity, and random heterogeneity. These cases, which are fully discussed in a few monographs,^[3-5] are briefly commented on here.

Nearly homogeneous surfaces

It may happen (for instance, in gas adsorption on carbon blacks) that the experimental data can be reproduced quite accurately by an equation that holds for ideal surfaces, but with parameters that do not coincide with those characteristic of the ideal surfaces. This case can be interpreted in terms of small deviations from ideality due to the defects (grain borders, pointlike or extended defects, etc.) unavoidably present at any real surface.

Random heterogeneous surfaces

The opposite case of quasi-homogeneity is that of random heterogeneity, in which the adsorption sites having the same value of the adsorption energy are assumed to be distributed completely at random on the surface. This situation, proposed to model equilibrium adsorption by Hill and Tompkins in the late 1940s, is however difficult to use and has had few practical applications.

Patchwise heterogeneous surfaces

Somewhat intermediate between quasi-homogeneity and random heterogeneity is the case of patchwise heterogeneity. One speaks of patchwise heterogeneity when the surface can be viewed as a collection of many energetically homogeneous patches, each large with respect to the atomic size. These patches (called *homotattic*) can thus be considered as indefinitely extended, and lateral interactions between molecules adsorbed on different patches can be ignored. The homotattic patch approximation, likely introduced for the first time by Sanford and Ross, consists of using for each patch the theory developed for the homogeneous surface.

Geometric Heterogeneity

Even surfaces obtained by cleaving single crystals are not atomically smooth, because the surface undergoes relaxation and reconstruction and hosts equilibrium defects like vacancies and self-interstitials. Roughening is a fundamental process for all solid surfaces and results in terraces, steps, and kinks (see, for instance, Bernasconi and Tosatti^[6] and Lapujoulade.^[7])

This situation is magnified for highly dispersed solids, which are characterized by strongly nonequilibrium configurations obtained by imparting (via fracture, supercritical extraction of solvent, burning of templates, etc.) a macroscopic energy over relatively few degrees of freedom. Highly dispersed solids are characterized by a fraction of atoms exposed to the atmosphere, which frequently is on the order of 10% or even higher. They provide examples of systems in which the surface properties do not disappear even in the thermodynamic limit. The techniques that are usually adopted for the characterization of flat surfaces (like low- or high-energy electron diffraction, scanning or transmission electron microscopy, atomic-force or scanning tunnel microscopy) are therefore unsuitable for the characterization of highly dispersed solids. Rather, the high number of atoms exposed to the atmosphere suggests the use of adsorption techniques for their characterization. The measurement of the monolayer amount of adsorbates with the same nature but scaled cross sections (e.g., noble gases, alkanes, or polymers) is one such method.

The physisorption of small molecules is usually accounted for in terms of the one-molecule/one-site assumption, stating that the adsorbate at the monolayer is so arranged that each adsorbed molecule covers one and only one adsorption site. This assumption is manifestly false for large molecules, which can behave as steric constraint to the adsorption on other sites even when they are attached to the surface through one site only.

An estimate of the adsorbent area, A , is obtained by multiplying the monolayer amount, N_m , by the adsorbate cross section, σ : $A = N_m \sigma$. A typical experimental arrangement is based on N_2 adsorption at 77 K; the surface area A° , determined by measuring in these conditions the monolayer amount N_m° and multiplying it by the N_2 cross section σ° (with $\sigma^\circ := \sigma_{N_2} = 16.2 \text{ \AA}^2$; Jelinek and sz. Kováts^[8] for a discussion of this value), is usually referred to as *standard surface area*, $A^\circ = N_m^\circ \sigma^\circ$.

The standard surface area provides an adequate description of the true surface area only if it remains unchanged when the probe is changed, i.e., if

$$N_m = N_m^\circ \sigma^\circ / \sigma. \quad (6)$$

The cross section σ depends on the number of atoms forming the molecule and on their spatial arrangement. Because this arrangement is modified by the vicinity of other molecules, σ is expected to depend on the adsorbent, too. Cerofolini et al.^[9] discuss a situation that seems to demonstrate the dependence of σ on the adsorbent. Though an unambiguous determination of the molecular cross section is a difficult, likely ill-posed, problem, extended compilations are available, based either on bulk (liquid or gas) phase properties of the adsorbate^[10] or on molecular dynamics simulations.^[11]

Starting from these compilations, one usually finds that even considering different adsorption probes with similar chemical nature (to minimize the differences in adsorption forces), the determined monolayer amount $N_m(\sigma)$ and area depend on the probe, in a way for which condition 6 is not satisfied. Two situations are typically encountered: A increases with σ , or A decreases with σ .

The area increases with adsorbate cross section—Smooth surfaces

This case is the one most frequently observed in practice. According to the adsorbate size, λ ($\simeq \sigma^{1/2}$), two extreme situations are observed. When λ is much smaller than the mean nearest-neighbor distance, ξ , between adsorption sites,

$$\lambda \ll \xi, \quad (7)$$

the measured monolayer amount is independent of σ , so that

$$A \propto \sigma. \quad (8)$$

Because typically $\xi \simeq 3 \text{ \AA}$, only a few adsorbates are such to guarantee that condition 7 is indeed satisfied; the most noticeable ones are He, H_2 (both requiring freezing in the cryogenic regime), and, marginally, Ne. Eq. 8 shows that these adsorbates do not provide a complete coverage of the surface.

For $\lambda \gg \xi$ (which happens for most adsorbates), the measured monolayer amounts vary with σ as

$$A = A^* + \delta(\sigma) \quad (9)$$

where $\delta(\sigma)$ is a slowly variable, monotonically increasing function such that $\delta(0) = 0$. Eq. 9 suggests that in this case the monolayer amount provides a complete coverage of the surface, the quantity $\delta(\sigma)$ being related to a partial adsorbate-adsorbate overlapping.

The area decreases with adsorbate cross section—Fractal surfaces

The case of A decreasing with σ is less frequently encountered in practice. In these cases, the monolayer amount is usually found to vary as

$$N_m = N_m^\circ (\sigma^\circ / \sigma)^{D/2}, \quad (10)$$

where D is a constant exponent ($D > 2$). The corresponding area varies as

$$A = A^\circ (\sigma^\circ / \sigma)^{D/2-1}. \quad (11)$$

Of course, Eqs. 10 and 11 have a quite restricted range-of-validity: these equations are satisfied for σ usually varying over less than one order of magnitude. Assuming their validity even in the limit for $\sigma \rightarrow 0$, they state that the area diverges in this limit. In abstract analysis, any surface satisfying condition 10 is said to be a fractal with Hausdorff fractal dimension D ; this suggests that real surfaces satisfying Eqs. 10 and 11 may be considered *fractal*, at least in the length scale where the preceding equations are actually observed experimentally.

The problem of the *experimental* determination of the fractal dimension D of real surfaces was first addressed by Pfeifer and Avnir,^[12] who proposed four different methods based on adsorption data. In a subsequent paper, Avnir, Farin, and Pfeifer applied those methods to real adsorbents and gave evidence for surfaces with $D > 2$.^[13]

A classification

In conclusion, real surfaces display many pathological behaviors. According to its fractal dimension D , a

surface is classified either as *smooth* when $D = 2$ or as *fractal* when $2 < D \leq 3$. Most adsorbents are smooth, though fractal adsorbents are occasionally found. Smooth surfaces, in turn, may be either *flat* (e.g., lamellar solids) or *rough* (e.g., zeolites, etc.). Rough adsorbents can often be viewed as a collection of convex and concave regions. The concave regions are usually referred to as *pores*.

Induced Heterogeneity

The heterogeneity just considered is essentially a property of the adsorbent–adsorbate pair in the zero-coverage limit. As adsorption proceeds, the situation is complicated by other facts, like the interaction between adsorbed molecules or the possibility that the adsorbent undergoes reconstruction. Both cases can be seen as resulting in an induced heterogeneity, which in general will impact on both energetic and geometric heterogeneity. However, it is customarily assumed that lateral interactions between adsorbed molecules are adequately described in terms of energy heterogeneity, while the surface reconstruction undergone during the adsorption–desorption process can be described in terms of geometric heterogeneity.

Induced energy heterogeneity

In the low-density limit, each adsorbed molecule can be assumed to feel the mean field produced by the remaining ones. In this case, the adsorption energy varies linearly with the density of the adsorbed layer, i.e., with the coverage. Actually, for a heterogeneous surface one must distinguish the average coverage, ϑ , from the local coverage, Θ . Upon which coverage the induced adsorption energy depends is specified by the nature of the fixed heterogeneity: q will vary linearly with ϑ for random heterogeneous surfaces, or with Θ for nearly homogeneous as well for patchwise heterogeneous surfaces.

Allostericity

An allosteric molecule is characterized by the presence of n equivalent sites, which modify their structure (binding energy) when the sites are progressively filled. Allosteric transitions play an important role in biology: The transport of oxygen by hemoglobin is a classic and important example.^[14] The theory of allostericity has been the matter of a seminal work (by Monod, Wyman and Changeux) of molecular biology.^[15]

Allostericity plays a major role not only in O_2 adsorption on hemoglobin, but also in other phenomena of large biological relevance, like the adsorption of polar molecules on proteins or their denaturation in suitable solvents. The description in this case is complicated by the wide variety of adsorption sites and of protein

configurations resulting after adsorption. In this case, the adsorption energy is comparable to the energy stabilizing the secondary structure of the protein, so that a kind of surface reconstruction takes place after the adsorption; accordingly, the adsorbent and adsorbate cannot be considered as slightly coupled entities, but rather as a new molecular system. Allostericity and heterogeneity combine together in a truly intriguing case in the adsorption of polar molecules on proteins.^[16]

Induced geometric heterogeneity

The case of induced geometric heterogeneity is associated with the progressive exposure of new sites while adsorption proceeds. It is certainly involved in protein unfolding during adsorption of polar molecules or exposure to a denaturing agent. A theory able to describe the adsorption under nearly equilibrium conditions for a surface undergoing surface reconstruction was proposed by Cerofolini.^[17]

Even adsorption in concave regions can be regarded as a kind of induced geometric heterogeneity. The induced heterogeneity is manifested when the thickness of the adsorbed film becomes comparable with the modulus of the curvature radius of the region. The major effect of this induced heterogeneity is the rise of hysteresis, characterized by the fact that while adsorption proceeds it occurs essentially in a layer-by-layer mode until the pore is filled and a liquid is formed. After that, adsorption and desorption are controlled by the meniscus inside the pore.

ADSORPTION–DESORPTION KINETICS

As discussed, adsorbent materials of practical interest are usually highly nonideal. It is nonetheless useful to consider which kinetics would be observed for adsorption–desorption phenomena occurring on ideal (homogeneous and flat) surfaces. In that case, it is convenient to describe the processes in terms of surface coverage, Θ .

The Rate Equation

The adsorption–desorption kinetics are usually expressed giving the time dependence of the surface coverage Θ . The rate equation is given by

$$N_s \frac{d\Theta}{dt} = J^+ - J^- \quad (12)$$

where N_s is the number of sites per unit area, while J^+ and J^- are the adsorption and desorption rate, respectively. The adsorption rate J^+ increases with the external pressure p , while the desorption rate J^- increases with the surface coverage Θ . Because $J^+ > 0$ for any $p > 0$

and $J^- > 0$ for any $\Theta > 0$, both J^+ and J^- should be considered in Eq. 12.

Experiments are, however, performed so as to have either $J^+ \gg J^-$ or $J^- \gg J^+$. The function $\Theta_+(t)$ satisfying Eq. 12 when the former condition is satisfied is referred to as *adsorption kinetics*, while the function $\Theta_-(t)$ satisfying Eq. 12 when the latter condition is satisfied is referred to as *desorption kinetics*.

In some cases, it is also possible to carry out the experiment in such a way as to have simultaneously $J^+ \simeq J^-$ and $|J^+ - J^-|/J^+ \ll 1$, varying slowly either p (at constant temperature) or T (at constant pressure). Provided that the rates of variation $\dot{p}/p$ or $\dot{T}/T$ are sufficiently low, the functions $\Theta_T(p)$ or $\Theta_p(T)$ can be assumed to be the *equilibrium isotherm* or *equilibrium isobar*, determined under quasistatic conditions.

The Rate Equation for Adsorption on Ideal Surfaces

The adsorption rate can formally be written as

$$J^+ = \kappa^+ \Phi^+, \quad (13)$$

where Φ^+ is the incoming flux and κ^+ is the probability that an incoming particle sticks at the surface.

The case of physisorption

The discussion of these terms is especially simple for physisorption, at least when this process is completed with the formation of one monolayer. In that case, indeed, Φ^+ is given by the kinetic theory of gases:

$$\Phi^+ = p / \sqrt{2\pi m k_B T}, \quad (14)$$

where m is the molecular mass and k_B is the Boltzmann constant. For small adsorbates (like noble gases or N_2), the coefficient κ^+ can simply be written as the product of the probability $(1 - \Theta)$ that the site is empty times the probability α that in a collision the molecule imparts to the adsorbent a sufficient energy to be captured in a bound state,

$$\kappa^+ = \alpha(1 - \Theta).$$

The sticking probability, α , is expected to depend on both T and Θ , $\alpha = \alpha(T, \Theta)$, because the kinetic energy before the impact depends on T and so does the phonon spectrum of the adsorbent, which in turn is modified to some extent by the presence of an adsorbate. The adsorption kinetics for physisorption are thus described by the rate equation

$$N_s \frac{d\Theta}{dt} = \alpha \frac{p}{\sqrt{2\pi m k_B T}} (1 - \Theta). \quad (15)$$

The case of chemisorption

Much more complex is the situation for chemisorption. Consider, for instance, the case of the hydrogenation by H_2 of carbon blacks. Because this process requires the cleavage of the strong H-H σ bond, it is thermodynamically allowed only in the presence of two close unsaturated carbon atoms able to form C-H bonds. This example shows that in chemisorption the site distribution plays an important role. If the reaction of the impinging molecule involves ν sites, the adsorption rate must contain a term, like $(1 - \Theta)^\nu$, describing their availability. For the determination of the other factors, we consider two particular cases: direct chemisorption and chemisorption via a physisorbed precursor.

Direct Chemisorption. Direct chemisorption is characterized by the reaction of the impinging gas-phase molecules with available adsorption sites. For direct chemisorption, the impinging flux is given by Eq. 14; the reaction is, however, possible only for sufficiently energetic molecules. This requirement is taken into account by assuming for κ^+ the following form:

$$\kappa^+ = \kappa_0^+ \exp(-E/\kappa_B T) (1 - \Theta)^\nu, \quad (16)$$

where E is the energy required to form the activated complex between the impinging molecule and the adsorption site and the preexponential factor κ_0^+ varies as a power law of T , so that it can be considered constant when compared to $\exp(-E/k_B T)$. The rate equation is thus given by

$$N_s \frac{d\Theta}{dt} = \kappa_0^+ \frac{p}{\sqrt{2\pi m k_B T}} \exp\left(-\frac{E}{k_B T}\right) (1 - \Theta)^\nu. \quad (17)$$

Chemisorption via a Physisorbed Precursor. Even in this case one has to consider the activation energy of the process through the factor 16. The impinging flux, however, is no longer taken from the kinetic theory of gases as in Eq. 14, but rather by

$$\Phi^+ = N_s \Theta_{ph} f_\perp,$$

where $f_\perp$ is the vibration frequency perpendicular to the surface and Θ_{ph} is the coverage of physisorption sites. This quantity may be estimated from its equilibrium value at the considered temperature and pressure, $\Theta_{ph}^{eq}(p/T)$. For chemisorption by a physisorbed precursor, the rate equation is thus given by

$$\frac{d\Theta}{dt} = (1 - \Theta)^\nu f_\perp \Theta_{ph}(p/T) \kappa_0^+ \exp\left(-\frac{E}{k_B T}\right). \quad (18)$$

Solving the rate equation

All the considered rate equations, namely, Eqs. 15, 17, and 18, have the form

$$\frac{d\Theta}{dt} = \frac{1}{\tau^+} (1 - \Theta)^v, \quad (19)$$

with τ^+ depending on the considered process:

- $\tau^+ = N_s \sqrt{2\pi m k_B T} / \alpha p$ for physisorption,
- $\tau^+ = N_s \sqrt{2\pi m k_B T} \exp(E/k_B T) / \kappa_0^+ p$ for direct chemisorption, and
- $\tau^+ = \exp(E/k_B T) / f_{\perp} \Theta_{ph}(p|T) \kappa_0^+$ for chemisorption from a physisorbed precursor.

In principle, the characteristic time, τ^+ , depends on Θ , because in principle so behave α , $f_{\perp}$, and E . In view of the difficulties in predicting such dependences, we shall ignore them (see, however, later), assuming for all the quantities their average values in the explored coverage interval. It will be assumed that Eq. 19 describes, for suitable choice of v and τ^+ (both independent of Θ), the adsorption kinetics on ideal surfaces in all cases.

If τ^+ is independent of Θ , Eq. 19 is immediately integrated by separation of variables

$$\int_{\Theta_0}^{\Theta} \frac{d\Theta}{(1 - \Theta)^v} = \frac{t}{\tau^+}, \quad (20)$$

where Θ_0 is the initial coverage.

Defining the reaction extent $\Delta^+(t)$ in such a way as to have $\Delta^+(0) = 0$ and $\Delta^+(+\infty) = 1$,

$$\Delta_v^+(t) := \frac{\Theta - \Theta_0}{1 - \Theta_0}$$

and integrating Eq. 20, the reaction extent is given by

$$\Delta_v^+(t) = 1 - \exp(-t/\tau^+) \quad (21)$$

for $v = 1$, or by

$$\Delta_2^+(t) = \frac{t}{t + \tau^+/(1 - \Theta_0)} \quad (22)$$

for $v = 2$.

The Rate Equation for Desorption from Ideal Surfaces

The discussion for desorption kinetics is much more compact than for adsorption kinetics. In fact, it is not necessary to consider separately either physisorption from chemisorption or the different chemisorption cases. The desorption flux can indeed be written as

$$J^- = (N_s f_{\perp} / v) \exp(-E/k_B T) \Theta^v. \quad (23)$$

What makes chemisorption different from physisorption are simply the values of E (typically higher than 1 eV for chemisorption or lower than 0.3 eV for physisorption) and of v ($v = 1$ for physisorption, while often $v > 1$ for chemisorption). Inserting Eq. 23 into Eq. 12 one gets

$$\frac{d\Theta}{dt} = \frac{\Theta^v}{\tau^-}, \quad (24)$$

with

$$\tau^- = \tau_0^- \exp\left(\frac{E}{k_B T}\right) \quad (25)$$

and $\tau_0^- := v/f_{\perp}$. Eq. 24 is immediately integrated by separation of variables

$$-\int_{\Theta_0}^{\Theta} \frac{d\Theta}{\Theta^v} = \frac{t}{\tau^-}, \quad (26)$$

where Θ_0 is the initial coverage.

Equation 24 can be solved in closed form both for first-order and second-order kinetics. In fact, defining the reaction extent

$$\Delta_v^-(t) := \frac{\Theta_0 - \Theta_v(t)}{\Theta_0},$$

one has

$$\Delta_1^-(t) = 1 - \exp(-t/\tau^-) \quad (27)$$

for first-order kinetics, and

$$\Delta_2^-(t) = \frac{t}{t + \tau^-/\Theta_0} \quad (28)$$

for second-order kinetics.

Irrespective of the reaction order, the reaction extent for desorption has the same formal expression as for adsorption; because of this we shall henceforth omit the upper indices “ $\pm$ ” specifying the case. The actual case will be clear from the context.

Accounting for Nonidealities

The adsorption–desorption kinetics are seldom described by the equations given in the previous part. Actually, all the nonidealities that affect the experiment (experimental protocol, adsorbent and adsorbate) influence the kinetics too.

The nonidealities of the protocol are associated with unavoidable desorption during adsorption experiments (especially sensitive at high coverage) or readsorption during desorption experiments (especially sensitive at low coverage). A theory for dealing with these

phenomena, the statistical rate theory, has recently been developed by Elliot and Ward.^[18,19]

The adsorbate nonidealities are essentially related to the possibility that internal degrees of freedom of the adsorbate are excited during adsorption. Though this process is intrinsically related to the catalytic properties of the surfaces, its description in formal kinetics has remained largely ignored. An example of such effects is discussed in Cerofolini, Meda, and Bandosz.^[9]

This entry, however, is primarily interested in understanding the effect of adsorbent nonidealities on the adsorption–desorption kinetics. According to the classification of the previous section, we consider geometric heterogeneity and energy heterogeneity.

The effects of geometric heterogeneity are manifold and partially unexplored. They span from a local increase of the gas density in the vicinity of the adsorbent surface (due to the superimposition of potentials generated by the walls defining concave regions) to the loss of concepts like “impinging flux” when one considers fractal surfaces. No general theory for dealing with such nonidealities has hitherto been developed.

The situation changes for energy heterogeneity, for which two general schemes have been proposed.

Induced energy heterogeneity in adsorption–desorption kinetics

Allowing for the effect of induced heterogeneity on adsorption–desorption kinetics is formally trivial. It suffices that one considers how E depends on Θ and inserts the function $E(\Theta)$ in Eq. 17 for direct chemisorption, Eq. 18, for chemisorption from a physisorbed precursor, or Eqs. 24 and 25 for desorption. The resulting equations can thus be solved by quadratures via separation of variables.

What turns out to be difficult is specifying the function $E(\Theta)$. If the forces attracting the molecule to the surface (“vertical” forces) are not exceedingly larger than the ones exerted among adsorbed molecules (“lateral” forces), then the lateral interactions can be described in the mean field approximation, so as to have

$$E(\Theta) = E_0 + \omega\Theta, \quad (29)$$

where ω is a parameter giving the strength of lateral interactions and $E_0 = E(0)$. It is noted that ω represents also the entire spread of adsorption energies felt during the process from $\Theta = 0$ to $\Theta = 1$.

There are, however, situations (essentially related to the case of adsorption of light molecules in micropores, in which the vertical forces are much stronger than the lateral ones) that cannot be described by a linear variation of E with Θ . Cerofolini et al.^[20] provides an exotic example of induced energy heterogeneity not described by Eq. 29.

Fixed energy heterogeneity on desorption kinetics

To some extent, the following description applies to both adsorption and desorption kinetics. In practice, however, it is applied to desorption kinetics only.

The desorption kinetics from patchwise heterogeneous surfaces are given by

$$\delta_v(t) = \int_{\mathcal{D}_\tau} \Delta_v(t, \tau) \phi(\tau) d\tau, \quad (30)$$

where $\delta_v(t)$ is the overall reaction extent at time t , $\phi(\tau)$ $d\tau$ is the fraction of surface with characteristic time between τ and $\tau + d\tau$, and $\mathcal{D}_\tau$ is the support of $\phi(\tau)$. Equation 30 originates the problem of extracting the distribution function $\phi(\tau)$ when the overall kinetics $\delta_v(t)$ are experimentally known.

The overall amount $\delta_v(t)$ depends on the lifetime distribution function $\phi(\tau)$ and on the set of initial conditions $\Theta_0(\tau)$. Clearly enough, if one wants to extract from the experimental function $\delta_v(t)$ the distribution function $\phi(\tau)$, one must operate in conditions for which $\Theta_0(\tau)$ is known. In the forthcoming discussion attention will be limited to the case of Θ_0 constant with τ . In this case, one may, without loss of generality, take $\Theta_0 = 1$, so that Eqs. 27 and 28 become, respectively,

$$\Delta_1(t, \tau) = 1 - \exp(-t/\tau) \quad (31)$$

and

$$\Delta_2(t, \tau) = \frac{t}{t + \tau}. \quad (32)$$

The theory proceeds more easily by specifying the desorption time τ as in 25. Because in most cases $\tau_0 \approx 10^{-13}$ s, the unique random variable characterizing the surface in desorption experiments is E . Rather than describing the surface with the distribution function $\phi(\tau)$, it is convenient to describe the surface with the distribution function $\phi(E)$, where $\phi(E)dE$ is the fraction of surface with desorption energy between E and $E + dE$. Because in the desorption kinetics of physisorbed molecules one may reasonably assume that E coincides with the adsorption energy q , and because $\phi(q)$ can be determined from adsorption equilibrium isotherms, there is the possibility of comparing the results of the kinetic theory with those of equilibrium theory.

Defining $\hat{\Delta}_v(t, E) := \Delta_v[t, \tau(E)]$, where $\tau(E)$ is given by Eq. 25, Eq. 30 becomes

$$\delta_v(t) = \int_0^{+\infty} \hat{\Delta}_v(t, E) \phi(E) dE. \quad (33)$$

Eq. 33 is specialized to the form

$$\delta_1(t) = \int_0^{+\infty} \left\{ 1 - \exp\left(-\frac{t}{\tau_0} \exp\left(-\frac{E}{k_B T}\right)\right) \right\} \varphi(E) dE \quad (34)$$

for first-order kinetics, and

$$\delta_2(t) = \int_0^{+\infty} \frac{1}{t + \tau_0 \exp(E/k_B T)} \varphi(E) dE \quad (35)$$

for second-order kinetics. Again, though the integration interval has been taken $(0, +\infty)$, in practice the support $\mathcal{D}_E$ of $\varphi(E)$ is contained in a finite interval (E_m, E_M) .

The integral representations 34 and 35 are the basic equations of the model. Substituting pressure p for time t , and adsorption energy q for activation energy E , Eqs. 34 and 35 resemble strictly the Langmuir or Jovanovic isotherm, respectively, of equilibrium adsorption.^[3,4,21] The analogy, however, is not complete: In fact, the equilibrium equations depend on $p \exp(q/k_B T)$, while the desorption kinetics depend on $t \exp(-E/k_B T)$ (note the different signs of the exponents). In spite of this, many of the methods developed for the theory of adsorption equilibrium on heterogeneous surfaces can be extended to desorption kinetics from heterogeneous surfaces.

For both the Langmuir and Jovanovic isotherms, several exact methods exist for the determination of the energy distribution function $\varphi(q)$ when the overall adsorption isotherm is precisely known. Rather than trying to modify these exact methods for desorption kinetics, attention will be concentrated on the condensation approximation (CA), which gives a simple analytical expression for the distribution function in terms of the derivative of the overall isotherm without producing unstable solutions.^[22,23] This approximation was introduced in the study of equilibrium adsorption on heterogeneous surfaces by Roginsky in the 1940. However, it was only after the Harris rediscovery and Cerofolini systematic applications to realistic model isotherms (more than 20 years later) that CA has become of wide use in the analysis of real adsorption systems. A well-documented history (with 504 references) of the theory of gas adsorption is given in Dabrowski.^[24]

The CA consists of replacing the kernel $\Delta_v(t, E)$ by the step function

$$\Delta_v^c(t, E) = \begin{cases} 1 & \text{for } t \geq t_v^c(E) \\ 0 & \text{for } t < t_v^c(E) \end{cases} \quad (36)$$

where the function $t_v^c(E)$ is determined by minimizing the distance d between the local kinetics $\hat{\Delta}_v(t, E)$ and their approximants in Eq. 36:

$$d[\hat{\Delta}_v(t, E), \Delta_v^c(t, E)] = \min. \quad (37)$$

It is immediately verified that condition 37 is satisfied for the Lagrangian distance and for the L^2 distance by taking

$$t_v^c(E) = [v - 1 + \ln(3 - v)] \tau_0 \exp(E/k_B T), \quad (38)$$

which applies to first-order and second-order kinetics. Defining the inverse function of 38,

$$\varepsilon_v := k_B T \ln(t/[v - 1 + \ln(3 - v)] \tau_0),$$

and putting $\hat{\delta}_v(\varepsilon_v) := \delta_v(t(\varepsilon_v))$, one has

$$\hat{\delta}_v(\varepsilon_v) = \int_{-\infty}^{\varepsilon_v} \varphi_{vc}(E) dE. \quad (39)$$

Of course, the solution of Eq. 39 will in general differ from the solutions of Eqs. 34 or 35, and for this reason the distribution function $\varphi_{vc}(E)$ in Eq. 39 has been denoted with the index “c.” Equation 39 is immediately solved by a differentiation with respect to ε_v :

$$\varphi_{vc}(\varepsilon_v) = \partial \hat{\delta}_v(\varepsilon_v) / \partial \varepsilon_v. \quad (40)$$

Because $\varepsilon_2 = \varepsilon_1 + k_B T \ln(\ln 2)$ and $d\varepsilon_2 = d\varepsilon_1$, the distribution function calculated in the hypothesis of first-order kinetics coincides with that calculated in the assumption of second-order kinetics, provided that the energy axis is shifted by an amount $-k_B T \ln(\ln 2) = 0.367 k_B T$.

As discussed in Cerofolini and Re,^[22,23] Eq. 40 gives an adequate description of the distribution function $\varphi(E)$ provided that the energy spectrum $\mathcal{D}_E$ is much wider than $k_B T$, $E_M - E_m \gg k_B T$ and $\varphi(E)$ varies slowly with E .

$$|\varphi'(E)|/\varphi(E) \ll 1/k_B T. \quad (41)$$

THE MOST FREQUENTLY OBSERVED KINETICS

Of the many isothermal kinetics observed on real surfaces for different processes like oxidation, chemisorption, and desorption, only two have deserved a special attention—the time-logarithm law and the time-power law.

The Elovich Equation

The time-logarithm law reads

$$\delta^E(t) = \begin{cases} \delta_m \ln(1 + t/t_m) & \text{for } 0 \leq t < t_M \\ \delta_M & \text{for } t_M \leq t \end{cases} \quad (42)$$

where δ_m , t_m , δ_M , and t_M are four characteristic parameters mutually related by the relationship

$$t_M := t_m [\exp(\delta_M/\delta_m) - 1], \quad (43)$$

t_M (and hence δ_M) may be allowed to go to $+\infty$. The macroscopic meaning of the parameters of Eq. 42 is straightforward: t_m is a characteristic time below which the logarithmic behavior is no longer observed, t_M is the time required to complete the process, while δ_m/t_m is the initial desorption rate.

As noticed in Landsberg^[25] and Rudzinski and Panczyk^[26] Eq. 42, though usually referred to as the “Elovich equation,” was actually reported by other authors. In fact, a time-logarithm law was first observed by Tamman and co-workers in 1922 for the *oxidation* of metals. Tamman’s work was rediscovered in 1935 by Vernon, who observed that the low-temperature oxidation of iron obeys Eq. 42. The oxidation of a number of metals and semiconductors has since then been reported to be described by Eq. 42; among them Cr, Mn, Fe, Co, Ni, Cu, Zn, Cd, Si, Sn, Pb, Sb, and Bi can be mentioned. The first observation of a *chemisorption* process obeying Eq. 42 was probably reported by Zel-dowitch in 1934, but it was only after its rediscovery by Elovich and co-workers that this equation became of wide use. On that basis the time-logarithm law 42 took the name of Elovich equation, though it could be more appropriately called the Zel-dowitch equation; it is also noted that in the German papers the person now commonly known as Elovich transliterated his name Elowitz.

A list of chemisorption systems obeying the Elovich equation is given in Aharoni and Tompkins.^[27] More recent, and not equally widespread, is the observation that *desorption* isotherms are often described by the Elovich equation.^[27]

The Time-Power Law

The desorption kinetics of many processes are described by a time-power law of the form

$$\theta^{\text{pl}}(t) = [1 + t/t_0]^{-s}, \quad (44)$$

where t_0 and s are temperature-dependent parameters, and t_0 is a characteristic time linked to the half-life time $t_{1/2}$ of the process by the relation $t_0 = t_{1/2}(2^{1/s} - 1)$. In the time domain $(0, +\infty)$, the kinetics described by Eq. 44 vary monotonically from $\theta = 1$ for $t = 0$ to $\theta \rightarrow 0$ for $t \rightarrow +\infty$. The reaction extent is given by $\delta^{\text{pl}}(t) = 1 - \theta^{\text{pl}}(t)$.

An example of process described by Eq. 44 is given by the low-temperature (below 200 K) recombination kinetics of CO to myoglobin (Mb, a heme group with a central iron atom) after photolysis of carboxymyoglobin MbCO.^[28,29] A short account of the kinetics that have been described by Eq. 44 is given in Landsberg.^[30]

The Folded–Unfolded Transition

Another situation of great biological interest is the folding–unfolding transition of macromolecules like nucleic acids and proteins. In particular, the unfolding of deoxynucleic acids (DNAs) is associated with the transcription of the information stored in the gene, while folding is associated with the quiet stage of the gene, in which it is particularly insensitive to its chemical environment. The folding–unfolding period determines the biological clock.

The folding degree of proteins, instead, is largely controlled by the medium. Most proteins are globular with fractal dimensions in the range 2.1–2.2, not much greater than the value $D = 2$ characteristic of smooth surfaces.^[31,32] Their biological functions are manifested in this configuration, and the adsorption–desorption phenomena they manifest are essentially associated with the formation of a layer of adsorbed solvent molecules. When the affinity of the amino acid residues with the environment is strong, however, the protein undergoes reconstruction with the exposure of new sites. This process seems unstable, so that above a certain unfolding the protein is denatured, thus losing its biological functions while assuming the conformation of a lightly reticulate polymer.

EXPLAINING THE MOST COMMON ADSORPTION KINETICS

Equation 42 or 44 cannot be described by the kinetics that are expected to occur on flat homogeneous surfaces. Rather, they are accounted for by assuming that the surface is energetically heterogeneous and possibly fractal.

Accounting for the Time-Logarithm Law

In view of its large occurrence in a lot of different situations, different explanations of the time-logarithm law have been proposed.

The first attempts to understand the time-logarithm law started immediately after its experimental discovery by Elovich himself. The first systematic search for the reasons explaining the Elovich equation was carried out by Porter and Tompkins almost 20 years later. These authors assumed that the time-logarithm law 42 is observed during activated chemisorption and interpreted the resulting equation in terms either of *preexisting surface heterogeneity* or of variation of the activation energy with coverage during the process (*induced heterogeneity*).

The following sections are devoted to discussing the picture resulting from assuming one or the other of such

extreme situations. It is also noted that the Elovich equation has also been analyzed in the light of the statistical rate theory.^[26]

Explaining the Elovich equation in terms of induced heterogeneity

In the induced-heterogeneity model the chemisorption process is assumed to be an activated process and the clean surface is considered as homogeneous with activation energy E_0 .

As the process progresses, the activation energy is then assumed to increase from E_0 to a value $E(\Theta)$, depending on the coverage, Θ . If one stipulates that chemisorption kinetics are described by the rate equation:

$$\frac{d\Theta}{dt} = r_E \exp\left(-\frac{E(\Theta)}{k_B T}\right) \quad (45)$$

(where r_E is a suitable kinetic coefficient independent of Θ) and E depends linearly on Θ as in Eq. 29, then one obtains a differential equation that can be integrated in closed form to give the time-logarithm behavior as in Eq. 42, provided that Θ is identified with δ/δ_M and

$$r_E \exp(-E_0/k_B T) = \delta_m/t_m, \quad (46)$$

$$\omega/k_B T = \ln(1 + t_m/t_m). \quad (47)$$

Though this derivation of the Elovich equation is very economical and succeeds in furnishing the empirical parameters δ_m and t_m with precise microscopic meaning, Eqs. 46 and 47, respectively, one cannot be completely satisfied with it because of the following reasons:

- It is not clear why the r.h.s. of Eq. 45 does not contain a factor like $(1-\Theta)^v$ that limits the chemisorbed amount for long adsorption time—see, for instance, Eq. 17;
- In many situations of practical interest the adsorbent is highly dispersed, which makes it difficult to accept that lateral interactions are responsible for the overall kinetics; and
- A linear dependence of E on Θ in the whole coverage range is undoubtedly a poor description of lateral interactions between chemisorbed molecules.

The last point deserves a special discussion. In fact, due to the short-range nature of the forces involved in chemisorption, $E(\Theta)$ is expected to depend on Θ only for $\Theta \gtrsim 1/c$, where c is the coordination number (i.e., the number of nearest neighbors) of each adsorption site. Because the Elovich behavior is associated with the final stages of the process, this is not a serious difficulty in activated chemisorption, where the appearance of the logarithmic kinetics is associated with high coverage.

The above chemisorption model can *formally* be extended to desorption kinetics. In fact, assuming the same rate equation as in Eq. 45,

$$-\frac{d\Theta}{dt} = r_E \exp\left(-\frac{E(\Theta)}{k_B T}\right)$$

(except, of course, for the sign of the derivative) and with a linear decrease of E with Θ , $E(\Theta) = E_0 - \omega\Theta$, one gets a differential equation whose solution is just the time-logarithm law. This derivation, however, is only formal because the logarithmic behavior is associated with the final stages of the desorption process where the coverage is negligible and E becomes independent of Θ .

Explaining the Elovich equation in terms of fixed heterogeneity

The time-logarithm kinetics observed in desorption are better explained in terms of fixed energy heterogeneity. Let us first consider second-order kinetics. Expressed in terms of ε_2 the Elovich Eq. 42 reads

$$\hat{\delta}_2^E(\varepsilon_2) = \begin{cases} \delta_m \ln\left(1 + \frac{\tau_0}{t_m} \exp\left(\frac{\varepsilon_2}{k_B T}\right)\right) & \text{for } -\infty < \varepsilon_2 \leq \varepsilon_{2M} \\ \delta_M & \text{for } \varepsilon_{2M} < \varepsilon_2 < +\infty \end{cases} \quad (48)$$

with

$$\varepsilon_{2M} := \varepsilon_{2m} + k_B T \ln[\exp(\delta_m/\delta_M) - 1]$$

and

$$\varepsilon_{2m} := k_B T \ln(t_m/\tau_0). \quad (49)$$

In the sequel, the index “2” to ε and φ_c will be omitted to have more manageable expressions. The application of Eqs. 40 to 48 gives

$$\varphi_c(\varepsilon) = \frac{\delta_m/\delta_M}{k_B T} \begin{cases} \exp\left(\frac{\varepsilon - \varepsilon_m}{k_B T}\right) / \left[1 + \exp\left(\frac{\varepsilon - \varepsilon_m}{k_B T}\right)\right] & \text{for } -\infty < \varepsilon \leq \varepsilon_M \\ 0 & \text{for } \varepsilon_M < \varepsilon < +\infty \end{cases} \quad (50)$$

whatever is the value of δ_M . A study of the distribution function 50 shows the following limiting behaviors:

$$\varphi_c(\varepsilon) \simeq \frac{\delta_m/\delta_M}{k_B T} \begin{cases} \exp\left(\frac{\varepsilon - \varepsilon_m}{k_B T}\right) & \text{for } -\infty < \varepsilon \leq \varepsilon_m - k_B T \\ 1 & \text{for } \varepsilon_m - k_B T \leq \varepsilon \leq \varepsilon_M \\ 0 & \text{for } \varepsilon_M < \varepsilon < +\infty \end{cases} \quad (51)$$

Of course, the calculated distribution function cannot be an arbitrary function of ε and must satisfy the mathematical conditions of non-negativity and of normalization; physical conditions too have to be satisfied. For instance, $\varphi_c(\varepsilon)$ must be identically null for $\varepsilon < 0$ and

must be temperature independent. There is no choice of ε_m and ε_M for which function 50 is temperature-independent; however, the asymptotic expansion 51 shows that $\varphi_c(\varepsilon)$ varies with T only in a region centered on ε_m of width $O(k_B T)$, while the calculated distribution function is approximately constant in the whole interval ($\varepsilon_m, \varepsilon_M$) provided that the extremes ε_m and ε_M , and therefore the difference $\varepsilon_M - \varepsilon_m$, are constant with T .

Equation 51 states that ε_M must be interpreted as the maximum adsorption energy, $\varepsilon_M = E_M$, while the distribution function vanishes exponentially with $\varepsilon - \varepsilon_m$ for $\varepsilon \leq \varepsilon_m$. Though in principle $\mathcal{D}_E$ does not admit a minimum, ε_m can in a way be interpreted as an estimate of the minimum adsorption energy, $\varepsilon_m \simeq E_m$.

Because in general $\varphi(E) \neq \varphi_c(E)$, even the maximum desorption energy E_M^c determined within the CA is in principle different from the true desorption energy E_M .

The normalization condition for $\varphi_c(\varepsilon)$,

$$\int_{-\infty}^{E_M^c} \varphi_c(\varepsilon) d\varepsilon = 1$$

gives the following equation for E_M^c :

$$\delta_m \ln \left(1 + \exp \left(\frac{E_M^c - \varepsilon_m}{k_B T} \right) \right) = \delta_M,$$

whose solution is

$$\begin{aligned} E_M^c - \varepsilon_m &= k_B T \ln \left(\exp \left(\frac{\delta_M}{\delta_m} \right) - 1 \right) \\ &\simeq k_B T \frac{\delta_M}{\delta_m}. \end{aligned} \quad (52)$$

Combining this relationship with the physical meaning of ε_m and ε_M , one eventually gets the meaning and temperature dependence of the parameters t_m and δ_m :

$$t_m \approx \tau_0 \exp(E_m/k_B T),$$

$$\delta_m \simeq \delta_M k_B T / \omega,$$

with $\varepsilon_m \simeq E_m$ and $\omega := E_M - E_m \simeq \varepsilon_M - \varepsilon_m$ being the width of the energy spectrum.

When first order kinetics are considered, the Elovich Eq. 42 reads

$$\hat{\delta}_1^E(\varepsilon_1) = \begin{cases} \delta_m \ln \left[1 + \frac{\tau_0 \ln 2}{t_m} \exp \left(\frac{\varepsilon_1}{k_B T} \right) \right] & \text{for } -\infty < \varepsilon_1 \leq \varepsilon_{1M} \\ \delta_M & \text{for } \varepsilon_{1M} < \varepsilon_1 < +\infty \end{cases} \quad (53)$$

with

$$\varepsilon_{1M} := \varepsilon_{1m} + k_B T \ln(\exp(\delta_M/\delta_m) - 1) \quad (54)$$

and

$$\varepsilon_{im} := k_B T \ln(t_m/\tau_0) - k_B T \ln(\ln 2). \quad (55)$$

The application of Eq. 40 to Eq. 53 gives a distribution function with the same shape as 50, in which ε_2 , ε_{2m} , and ε_{2M} are replaced by ε_1 , ε_{1m} , and ε_{1M} , respectively. All the considerations on minimum and maximum adsorption energies made above for second order kinetics therefore apply to first-order kinetics provided that ε_{2m} and ε_{2M} are replaced by ε_{1m} and ε_{1M} , respectively, as calculated from Eqs. 54 and 55.

Irrespective of the kinetic order, the CA is valid when the energy spectrum is much wider than $k_B T$; this condition is fulfilled only for $\delta_m/\delta_M \ll 1$, which is the condition for the validity of the preceding argument. Note that when this condition is satisfied, the difference between the calculated distribution functions for first- and second-order kinetics, consisting of a shift of the order of $k_B T$ on the energy axis, is negligible.

Combining this analysis with that of the previous section, it follows that *irrespective of the assumed theoretical framework (fixed or induced heterogeneity), the parameters δ_m and t_m have the same meaning*: Eqs. 47 and 49 state that t_m is essentially related to the minimum activation energy for chemisorption.

Explaining the Elovich equation in terms of surface reconstruction

While both the fixed-heterogeneity model and the induced-heterogeneity model ascribe the Elovich behavior to (preexisting or chemisorption-induced) energetic factors, the model proposed by Landsberg^[25] ascribes it to topographic factors related to surface reconstruction during chemisorption. The following description applies to chemisorption, but *mutatis mutandis* it can be extended to oxidation.

Landsberg postulated that

- (L1) Chemisorption occurs via the collision of impinging gas-phase molecules on free surface sites,
- (L2) Chemisorption is responsible for surface reconstruction via generation of new sites at a rate proportional to the chemisorption rate, and
- (L3) Each newly exposed zone has the same topography as the surface from which it was generated.

The Landsberg model can be formalized as follows. Assumption (L1) reads

$$\frac{dN}{dt} = N_{s^*} \sigma \alpha \Phi^+, \quad (56)$$

where N and N_{s^*} are the numbers of chemisorbed molecules and available chemisorption sites, respectively, both referred to as the unit geometric area. The models described earlier for adsorption kinetics on

homogeneous surfaces are reproduced by taking

$$\begin{aligned} N &= N_s \Theta \\ N_{s^*} &= N_s (1 - \Theta) \\ N_s &= \text{constant} \\ v_s &= 1 \end{aligned} \quad (57)$$

Landsberg, however, did not accept assumptions 57 and postulated (L2); in so doing he postulated that chemisorption has two contrasting effects: On one side the chemisorption of a molecule deactivates one site, but on another side it is also responsible for the creation of newly exposed area and hence of new adsorption sites. This can be restated by assuming that the active area lost per unit time and unit geometric area is $b \, dN/dt$, where b represents the net lost area per chemisorbed molecule. If the reconstruction mechanism is such that (L3) is satisfied, the number of available sites per unit geometric area evolves with time as

$$-\frac{dN_{s^*}}{dt} = b \frac{dN}{dt} N_{s^*} = \alpha \sigma b \Phi^+ N_{s^*}^2. \quad (58)$$

Equation 58 can be integrated by variable separation and gives

$$\frac{N_{s^*}(0)}{N_{s^*}(t)} = 1 + \Phi^+ N_{s^*}(0) \alpha \sigma b t, \quad (59)$$

where $N_{s^*}(0)$ is the density of surface sites at time $t = 0$. Inserting Eq. 59 into Eq. 56, one gets a differential equation that can be solved by variable separation:

$$N(t) = \frac{1}{b} \ln \left(1 + b \frac{t}{a} \right), \quad (60)$$

with $a = N_{s^*}(0) \alpha \sigma / \Phi^+$. The comparison of the time-logarithmic behavior in Eq. 42 with 60 shows that they coincide for $\delta_m = 1/bN_s$ and $t_m = a/b$. This comparison does, therefore, furnish the empirical parameters of the Elovich equation with a microscopic structural meaning.

In modern language, assumption (L3) states that the surface reconstructs in a kind of self-similar way. Though the exact mathematical specification of this self-similarity is an open problem, the appearance of the Elovich equation seems essentially related to a fractal nature of the reconstructed surface. This property is immediately understood by observing that the number of molecules accommodated on a given finite geometric surface area diverges as $t \rightarrow +\infty$.

The increase with time of the number of chemisorbed molecules is independent of b in the linear regime,

$$t \rightarrow 0 \Rightarrow N \sim \frac{1}{b} \frac{t}{t_m},$$

and varies inversely with b in the long-time regime,

$$t \rightarrow +\infty \Rightarrow N \sim \frac{1}{b} \ln \left(b \frac{t}{a} \right) \sim \frac{1}{b} \ln t.$$

Adsorption generates a surface whose area, measured with a molecular probe of assigned size, increases continuously as adsorption proceeds. The infinite surface area of a surface reconstructing according to (L1), (L2), and (L3) is actual (i.e., it can be measured with real molecular probes) though obtained only for $t \rightarrow +\infty$. The infinite surface area of common fractal adsorbents is potential (i.e., it results from an extrapolation process in the limit of null size of the adsorbed probe). The complexity of a surface reconstructing according to (L1), (L2), and (L3) is different than that of highly dispersed fractal adsorbents.

The swelling of the original surface resulting from chemisorption complicates the understanding of surface topography. Information on the topography of the reconstructed surface can be obtained by considering the region involved in chemisorption expressed in terms of original adsorbent. Let x be the thickness of the adsorbent region involved in the progressive increase with time of N . There are two possibilities: Either $N(t) + N_{s^*}(t)$ increases with t faster than $x(t)$ or not. In the latter case, the peak-to-peak roughness of the chemisorbed layer increases with t and the layer never reverts to a continuous homogeneous phase; otherwise, at a certain time $t_* - dt$, with t_* such that

$$N(t) + N_{s^*}(t) = \varrho x(t)$$

(where ϱ is the atomic density of the compound formed after chemisorption), the surface will attain its maximum geometric heterogeneity, while at $t = t_*$ the entire layer of thickness $x(t_*)$ will be constituted by the chemisorbed compound, thus producing a flat two-dimensional surface.

Accounting for the Time-Power Law

Equation 34 can manifestly be solved exactly via Laplace transform techniques. In the following, attention will, however, be limited to energy heterogeneity, which explains in the CA the time-power law.

Defining $\varepsilon := k_B T \ln (t/\tau_0 \ln 2)$, the time-power law 44 expressed in terms of ε reads

$$\hat{\delta}^{p1}(\varepsilon) = 1 - \left[1 + \frac{\tau_0 \ln 2}{t_0} \exp \left(\frac{\varepsilon}{k_B T} \right) \right]^{-s} \quad (61)$$

for $-\infty < \varepsilon < +\infty$. The straightforward application of Eq. 40 to Eq. 61 gives:

$$\varphi_c(\varepsilon) = \frac{s}{k_B T} \left[1 + \exp \left(\frac{\varepsilon - \varepsilon_0}{k_B T} \right) \right]^{-s-1} \exp \left(\frac{\varepsilon - \varepsilon_0}{k_B T} \right) \quad (62)$$

where $\varepsilon_0 := k_B T \ln(t_0/\tau_0 \ln 2)$. A study of 62 shows that $\varphi_c(\varepsilon)$ has a maximum in ε_0 and that

$$\varepsilon \ll \varepsilon_0 \Rightarrow \varphi_c(\varepsilon) \simeq \frac{s}{k_B T} \exp\left(\frac{\varepsilon - \varepsilon_0}{k_B T}\right),$$

$$\varepsilon_0 \ll \varepsilon \Rightarrow \varphi_c(\varepsilon) \simeq \frac{s}{k_B T} \exp\left(s \frac{\varepsilon_0 - \varepsilon}{k_B T}\right).$$

The distribution function $\varphi_c(\varepsilon)$ satisfies the mathematical conditions of nonnegativity and normalization. Moreover, though the distribution function 62 depends on temperature, this behavior may have a physical meaning, at least for adsorbents (like heme proteins) with many conformational substates. Above a given freezing temperature, these substates are in equilibrium and their population depends on the temperature according to a Boltzmann distribution.

It is mentioned, without further discussion, that the time-power law can be explained also by the Landsberg model, assuming a noninteger-order kinetics for the site-molecule reaction^[30] and that kinetics like Eq. 44 can be interpreted in terms of self-similar dynamics.^[33]

Accounting for the Folded–Unfolded Transition

Though Landsberg considered only the case $b > 0$, the case of negative b is interesting too and deserves an analysis. For $b < 0$, the solution of Eqs. 56 and 58 reads:^[34]

$$N(t) = -\frac{1}{|b|} \ln\left(1 - |b| \frac{t}{a}\right).$$

Thus, while for $b < 0$ the surface reconstruction leads to a smooth increase with t of the chemisorbed amount (in such a way that the number of chemisorbed molecules diverges only for $t \rightarrow +\infty$), for $b < 0$ the surface reconstruction is sudden and leads to a divergence of N at a finite time t_∞ , $t_\infty = a/|b|$. Though the case $b < 0$ may appear quite formal, the denaturation of proteins after exposure to a denaturing agent, or the fragmentation of the sphere like support of Ziegler–Natta catalysts produced during olefin polymerization in industrial heterogeneous processes, provides examples of Landsberg reconstruction with $b < 0$.^[34]

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